

Architecture

Continuum is a pnpm/Turborepo monorepo on Vercel, backed by Neon Postgres with pgvector. This describes what exists and why each boundary sits where it does.

Live: <https://continuumstudy.vercel.app>

Repository layout

```
apps/  
  web/           Next.js 15.5 App Router, React 19. The product.  
  obsidian-plugin/ Reads and writes the same workspace from an Obsidian vault.  
  video/         Remotion project for the demo film.  
packages/  
  ai/           Provider clients, routing policy, embeddings, health checks.  
  db/           Drizzle schema, migrations, seed and demo-seed.  
  domain/       Pure logic: mastery, spaced repetition, scheduling, permissions.  
  mcp/          MCP server definition: 46 tools with scopes and classes.  
  retrieval/    Vector and lexical retrieval primitives.  
  schemas/      Zod schemas shared across every boundary.
```

The split is not cosmetic. `packages/domain` holds every function that decides something about a learner, and it imports no database client, makes no network call, and calls no model. That is what makes the mastery model and the review scheduler testable as mathematics rather than as behaviour observed through three layers of I/O.

Data model

67 tables. The ones that carry the product's claims:

Table	Holds	Why it matters
<code>learning_states</code>	Four mastery dimensions per concept, plus <code>interval_days</code> , <code>ease</code> , <code>reps</code> , <code>lapses</code>	One row is both what a learner knows and when to show it again
<code>source_chunks</code>	Passage text, <code>content_hash</code> , <code>vector(1536)</code>	The unit a citation points at
<code>memory_chunks</code>	Durable memory with <code>importance</code> , <code>occurred_at</code> , <code>superseded</code>	What the assistant recalls across sessions
<code>claim_evidence</code>	Claim to chunk, with <code>status</code> and <code>verifier_route_id</code>	A research claim carries the passages that support or contradict it
<code>memory_events</code>	Append-only event log	Records are projections; events are the truth
<code>question_bank_attempts</code>	Per-attempt score and completion time	Feeds the review schedule
<code>assistant_sessions</code> / <code>assistant_messages</code>	Threads, pinning, archiving	Conversation history retrieval can read
<code>session_receipts</code>	What a session accomplished, with evidence ids	How an external agent hands work back

Both embedding columns are 1536-dimensional and indexed with HNSW over `vector_cosine_ops`. The dimension is asserted at write time:

```
if (values.some((value) => value.length !== dimensions))
  throw new Error(`Embedding model output must be ${dimensions}-dimensional to match the pgvector column`);
```

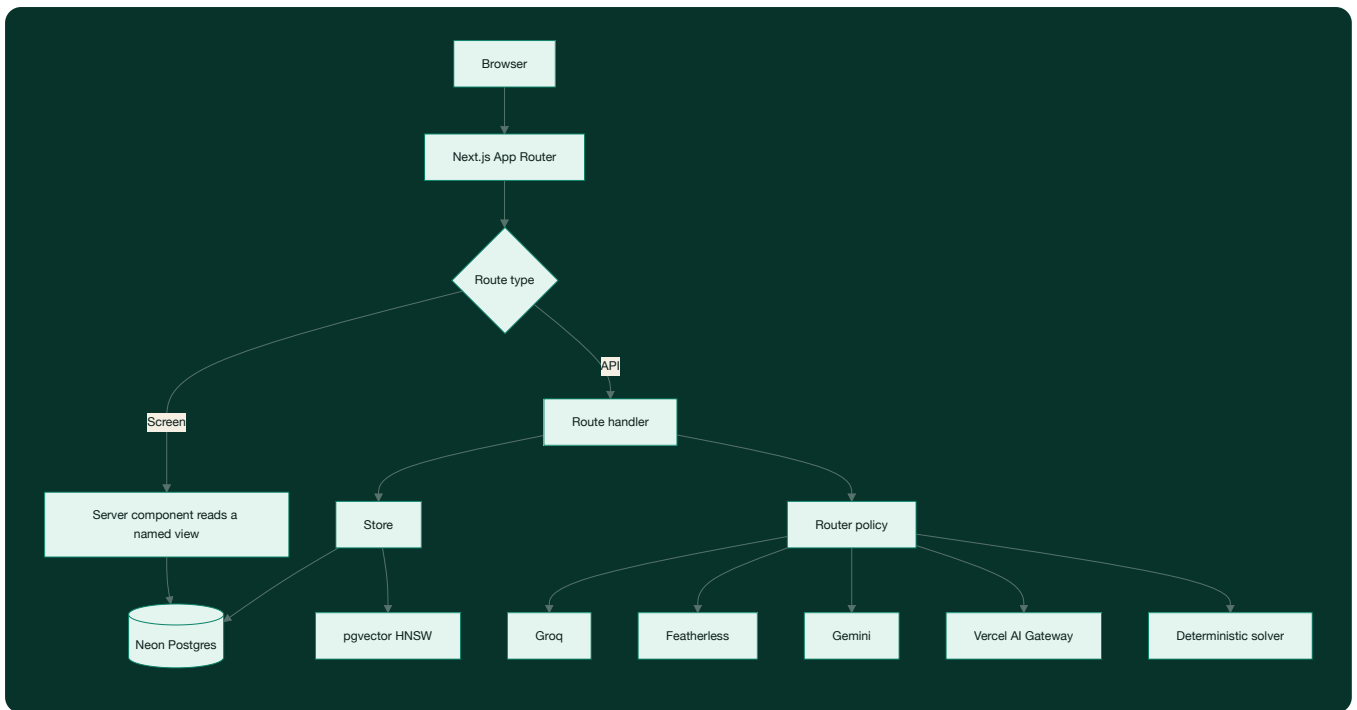
A provider swap that silently changes dimensionality is a failure that otherwise surfaces months later as quietly bad search results rather than as an error.

Events before records

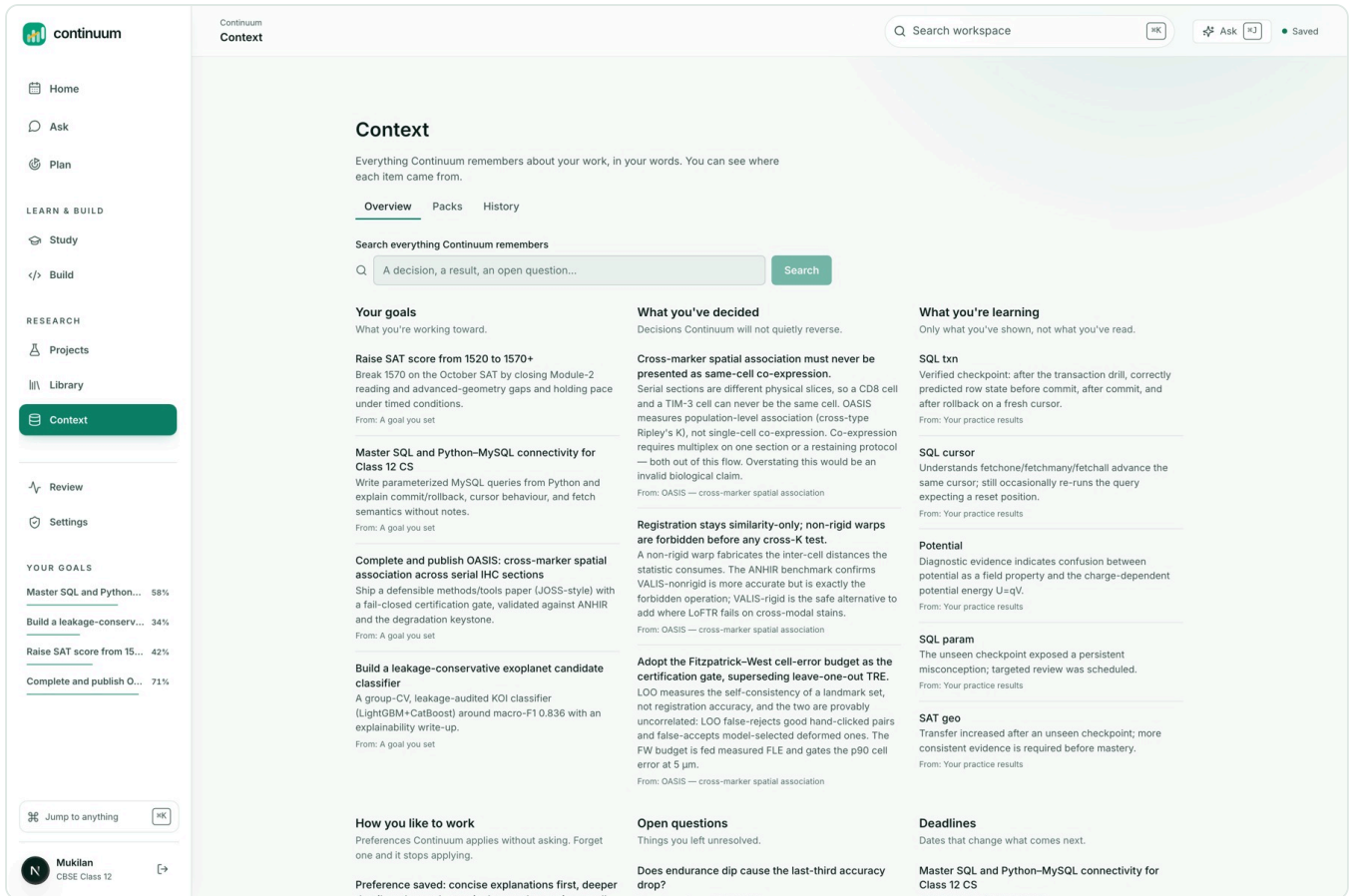
`memory_events` is append-only. `memory_records` are projections carrying a `superseded` flag, and `entity_summaries` hold an `event_watermark` so a summary knows how much history it has seen. Nothing is destructively updated. When the assistant proposes a change, the proposal is a row; approving it writes a new event rather than mutating the old one.

That is why the Review screen can show a field-level diff of every pending change, and why `whats_changed` can answer "what happened since this timestamp" without a dedicated changelog table.

Request path



51 API route handlers. Every screen reads through a named view rather than querying tables directly, and `tests/view-contract.test.ts` asserts a screen never reads a field its view does not return. That test exists because four panels once rendered permanently empty: the component asked for a field, the view never returned it, and an empty array renders as a well-designed "nothing here yet" which is indistinguishable from the truth.



Context. The routing table, provenance and token accounting the architecture below produces.

The store boundary

`apps/web/lib/store.ts` is the only thing that talks to the database on behalf of a request, and it exposes the same surface to three consumers:

1. The web app's server components and route handlers.
2. The MCP server, so Claude or any MCP client sees exactly the app's data.
3. The Obsidian plugin.

One implementation means an external agent cannot see a different workspace than the user does, and the scope system is enforced once. Every MCP tool declares a `requiredScope` and a `class` of `read` or `write`, and write tools route through a proposal rather than mutating directly:

```
tool({
  name: "save_progress_note",
  description: "... This cannot mark work complete: completion is a change the user approves in Continuum, so use propose_change for that.",
  requiredScope: "memory:write",
  class: "write",
})
```

Retrieval

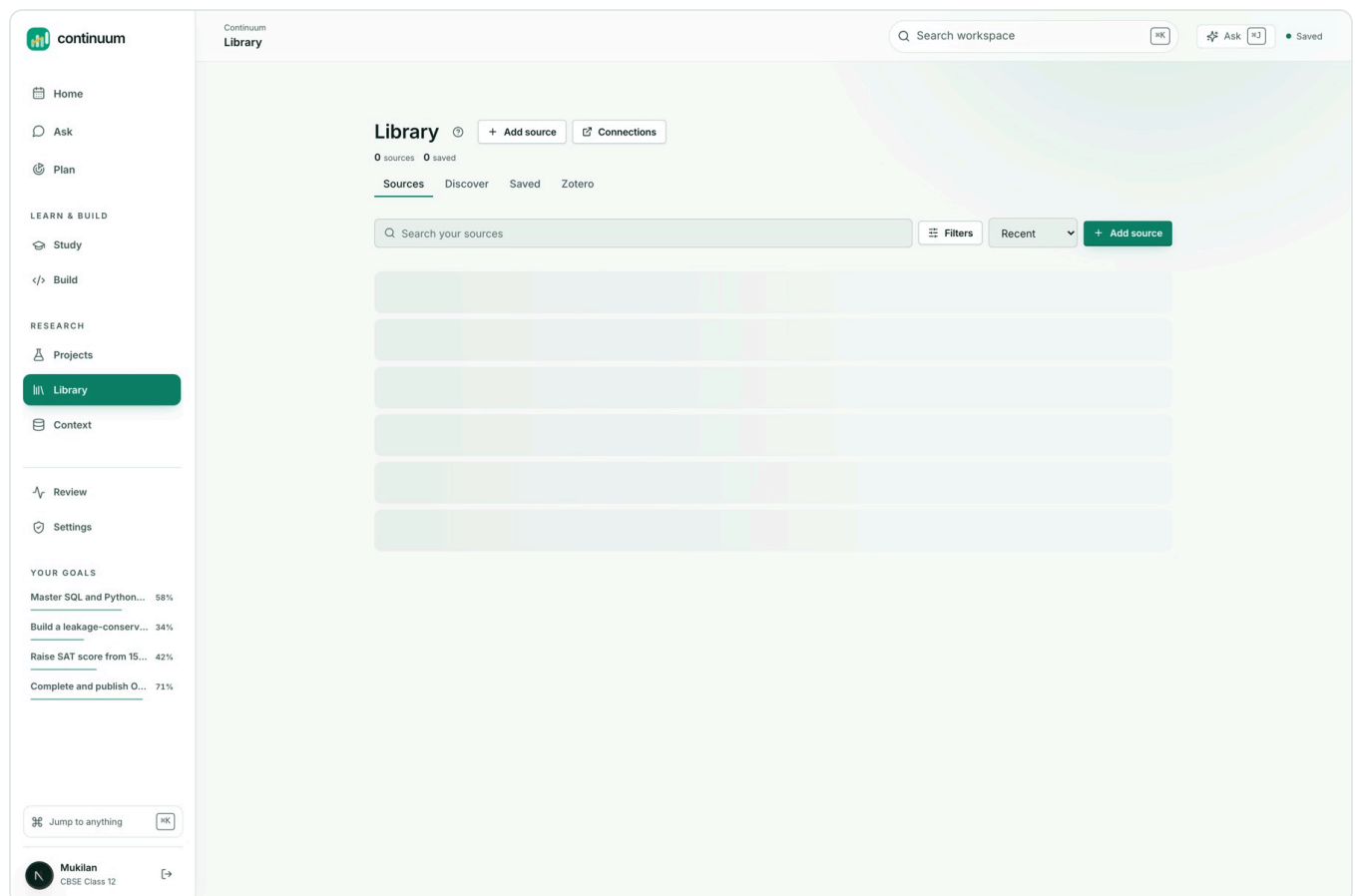
Two HNSW cosine indexes: `source_chunks.embedding` for imported documents, `memory_chunks.embedding` for durable memory.

Vector and lexical search race rather than lexical running on vector failure:

```
const [vectorHits, perTerm] = await Promise.all([vector, lexical]);
if (vectorHits.length) return vectorHits;
// round-robin across terms so one common word cannot fill the set alone
```

Awaiting the embedding and only falling back on an exception means a slow but successful embedding spends the whole deadline, the deadline fires, and the leg returns empty. That is a latency spike wearing the costume of an empty library. Racing makes the failure mode "slower but correct" rather than "fast and wrong".

`searchSourceChunksLexical` is a passage-only query. The general `searchResearch` concatenates claims, decisions, notes, and passages in that order and then slices, so asking it for six results on a term that also appears in a decision returns six decisions and zero passages. It was structurally incapable of returning the thing it was being used to return.



Library. Imported sources with their processing state, chunk counts and retention.

Deadlines and degradation

The assistant's four retrieval legs run concurrently under explicit deadlines:

```
export const DEADLINES = {
  classification: 1_500,
  retrieval: 2_000,
  sources: 3_500,
  pageContext: 300,
} as const;
```

A leg that misses its deadline contributes an empty array and appends to a `degraded` list that reaches the user as a disclosure. The assistant says "Answered from general knowledge, nothing in your workspace matched" rather than answering confidently from nothing. That disclosure is the only reason a five-link retrieval failure was ever findable, documented in `docs/retrieval-chain.md`.

Testing

1,117 unit and component tests across 71 files, plus Playwright end-to-end, accessibility, responsive, and visual specs.

Three test families are unusual enough to name:

`tests/view-contract.test.ts` asserts the screen-to-view contract; that the passage-only query exists and is what the assistant's lexical fallback calls; that read endpoints do not statically import write-path dependencies; and that every component owning a namespaced class family imports the stylesheet defining it. That last assertion exists because `concept-map.tsx` imported no stylesheet at all. Its rules live in `study.css`, so the concept map rendered correctly on `/learn` and `/study` and rendered as bare inline text on the goal page it was built for.

`tests/design-tokens.test.ts` holds `globals.css` under a 600-line ceiling and asserts every co-located stylesheet composes tokens rather than literal colours.

`tests/demo-seed.test.ts` asserts the seeded demo is internally consistent, for instance that the question-bank set scoring lowest is the one whose concept has the most lapses.

Deployment

Vercel, region `sin1`, Fluid Compute. Neon Postgres with pgvector. `sharp` and the PDF and DOCX parsers sit behind lazy loaders, because a static import failure in a serverless bundle takes down every export in the file. `sharp` failing to `dlopen` on linux-x64 once returned a 500 HTML page from `/api/sources`, which the Library rendered as "We couldn't load your sources".